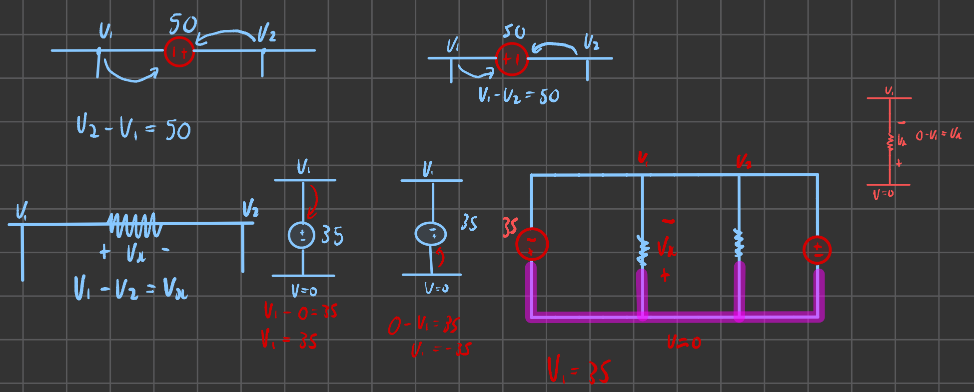
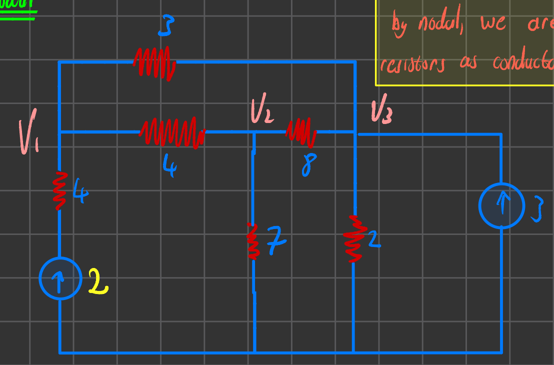
Recap



Inspection by Nodal

When we do inspection by nodal, we are using resistors as conductance ($G = \frac{1}{R}$)

V_1
$\boxed{} - 2 = 0$
$= 2$
V_2
$\boxed{} - 0 = 0$
V_3
$\boxed{} - 3 = 0$



$$\begin{array}{c|ccc} & 1 & 2 & 3 \\ \hline 1 & \frac{7}{12} & -\frac{1}{4} & -\frac{1}{3} \\ 2 & -\frac{1}{4} & \frac{24}{56} & -\frac{1}{8} \\ 3 & -\frac{1}{3} & -\frac{1}{8} & \frac{23}{24} \end{array} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ 3 \end{bmatrix}$$

G between V_1 & V_2 — G between V_1 & V_3 — V_2 & V_3

V_1

$\frac{1}{4} + \frac{1}{3}$

$\frac{V_1 - V_3}{3} + \frac{V_1 - V_2}{4} - 2 = 0$

$\frac{1}{4} + \frac{1}{7} + \frac{1}{8}$

$\frac{1}{8} + \frac{1}{3} + \frac{1}{2}$

Add all conductance attached to V_1
multiply upper lower triangle with negative

$$\begin{array}{c|ccc} & 1 & 2 & 3 \\ \hline 1 & R & -\frac{1}{4} & \theta \\ 2 & -\frac{1}{4} & \frac{24}{56} & -\frac{1}{8} \\ 3 & -\frac{1}{3} & k & \frac{23}{24} \end{array} \begin{bmatrix} V_1 \\ V_2 \\ V_3 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \\ P \end{bmatrix}$$

G between V_1 & V_2 — G between V_1 & V_3 — V_2 & V_3

$R = \frac{1}{4} + \frac{1}{3} = \frac{7}{12}$

4 below is not added, current is present

V_1

$\frac{V_1 - V_2}{4} + \frac{V_1 - V_3}{3} = 2$

$\frac{7V_1}{12} - \frac{V_2}{4} - \frac{V_3}{3} = 2$

$\frac{7}{12}V_1 - \frac{1}{4}V_2 - \frac{1}{3}V_3 = 2$

$\theta = -\frac{1}{3}$

symmetric

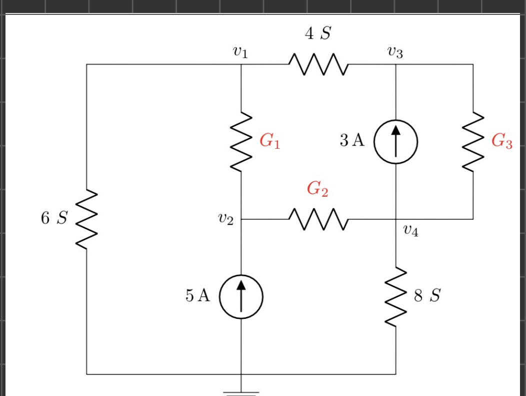
V_1 & V_3

$k = -\frac{1}{8}$

$P = 3$

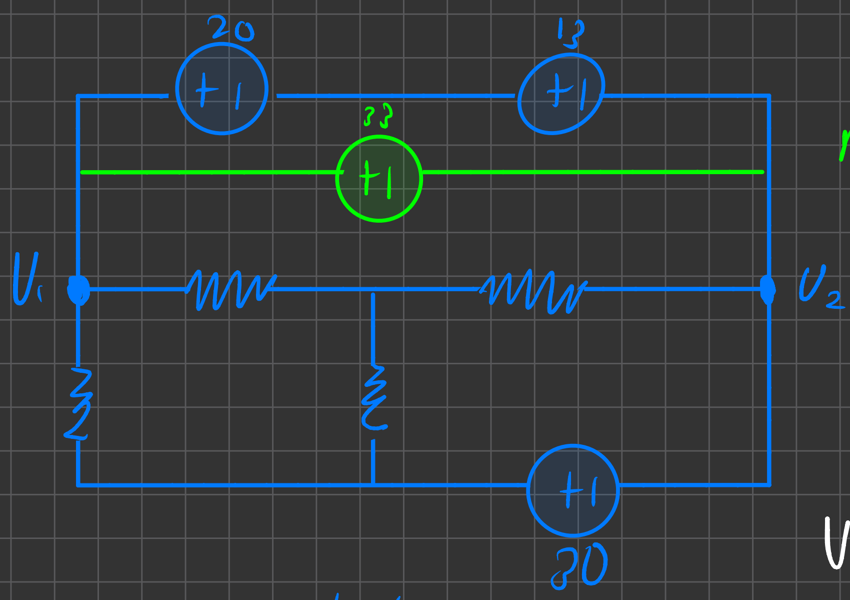
$\frac{V_2 - V_1}{4} + \frac{V_2}{7} + \frac{V_2 - V_3}{8} = 0$

$-\frac{1}{4}V_1 + \frac{24}{56}V_2 - \frac{1}{8} = 0$



v
$w = -G_2$
$x = G_3 + 8 + G_2$
$y = 3$

$G_1 = 10$
$G_2 = 6$
$G_3 = 9$
$G_1 + G_2 =$



new line

$V_2 = -30$

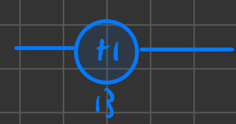
$V_1 - V_2 = 33$

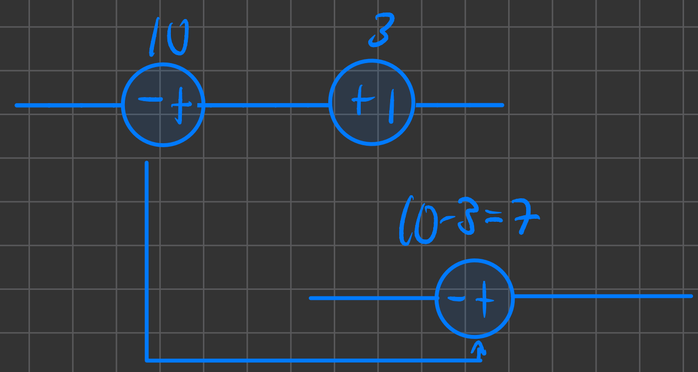
$V_1 - (-30) = 33$

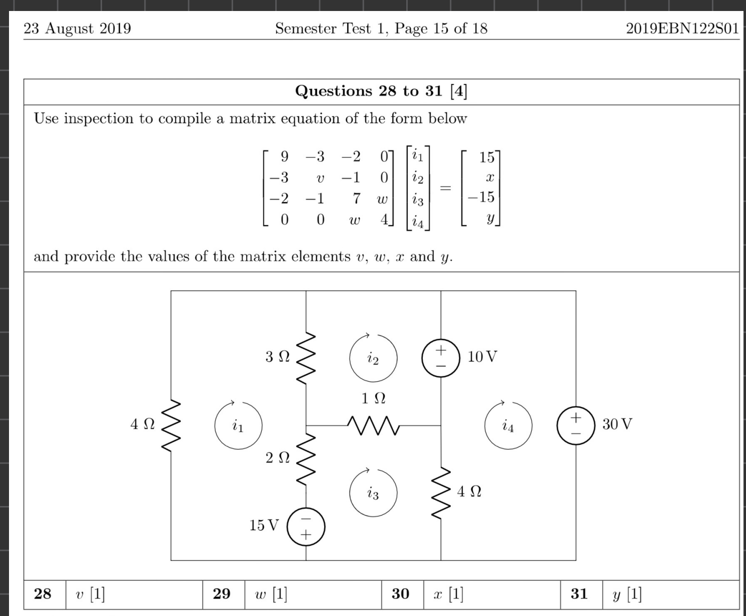
same polarity

take the larger value

and add the smaller value, maintaining polarity and shape of the bigger value







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Questions 28 to 31 [4]

Use inspection to compile a matrix equation of the form below

$$\begin{bmatrix} 9 & -3 & -2 & 0 \\ -3 & v & -1 & 0 \\ -2 & -1 & 7 & w \\ 0 & 0 & w & 4 \end{bmatrix} \begin{bmatrix} i_1 \\ i_2 \\ i_3 \\ i_4 \end{bmatrix} = \begin{bmatrix} 15 \\ x \\ -15 \\ y \end{bmatrix}$$

and provide the values of the matrix elements v, w, x and y.

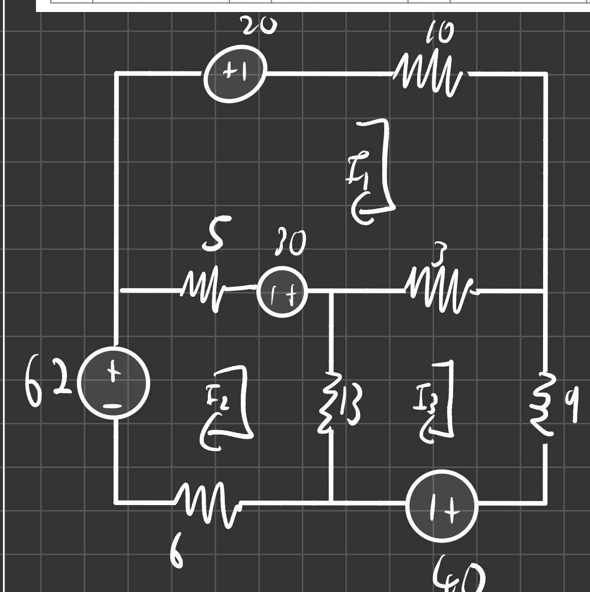
28	v [1]	29	w [1]	30	x [1]	31	y [1]

$v = 4$

$w = -4$

$x = -10$

$y = -20$



$$\begin{bmatrix} 18 & -5 & -3 \\ -5 & 24 & -13 \\ -3 & -13 & 25 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \\ I_3 \end{bmatrix} = \begin{bmatrix} -50 \\ 92 \\ -40 \end{bmatrix}$$